

MultiSenseNet: A unified approach to multi-task machine learning



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INTRODUCTION

BACKGROUND

Multi-task machine learning (ML) requires the design of ML models that are capable of processing diverse data types, such as images, audio, and text, while simultaneously handling multiple tasks. Achieving this requires integrating these tasks and data types into a unified framework. Despite an increasing number of models being developed to deal with this problem, there is currently no clear definition of a unified ML task and data representation. In this work, we aim to introduce the first of such definitions and present a novel architecture called MultiSenseNet (MSN) for processing these efficiently.

OBJECTIVES

- ❖ Define unified frameworks for arbitrary ML tasks and datasets.
- ❖ Compare the performance of different model architectures implemented.

POTENTIAL IMPACT

This work can advance the field of multi-task ML and enable a wide range of innovative applications across different domains. The application domains we are focused on in our research group are those related to climate change mitigation and adaptation.

METHODOLOGY

❖ **Data Collection and Cleaning:** We acquired, cleaned, and preprocessed publicly available datasets and organized them according to their respective tasks (Refer to Figure 1).

Dataset	Modality	ML task	Subtask
cat & dog images	image	supervised	image classification
audio collection	audio	supervised	speech recognition
food delivery	categorical, continuous	supervised	regression
corona sentiment	text, categorical	supervised	multi classification
various images	image	unsupervised	clustering

Fig. 1 Table showing collected datasets and their respective tasks.

❖ **Machine Learning Model:** Since we have various data modalities, we employed different pre-trained models in our multi-task architecture: MultiSenseNet.

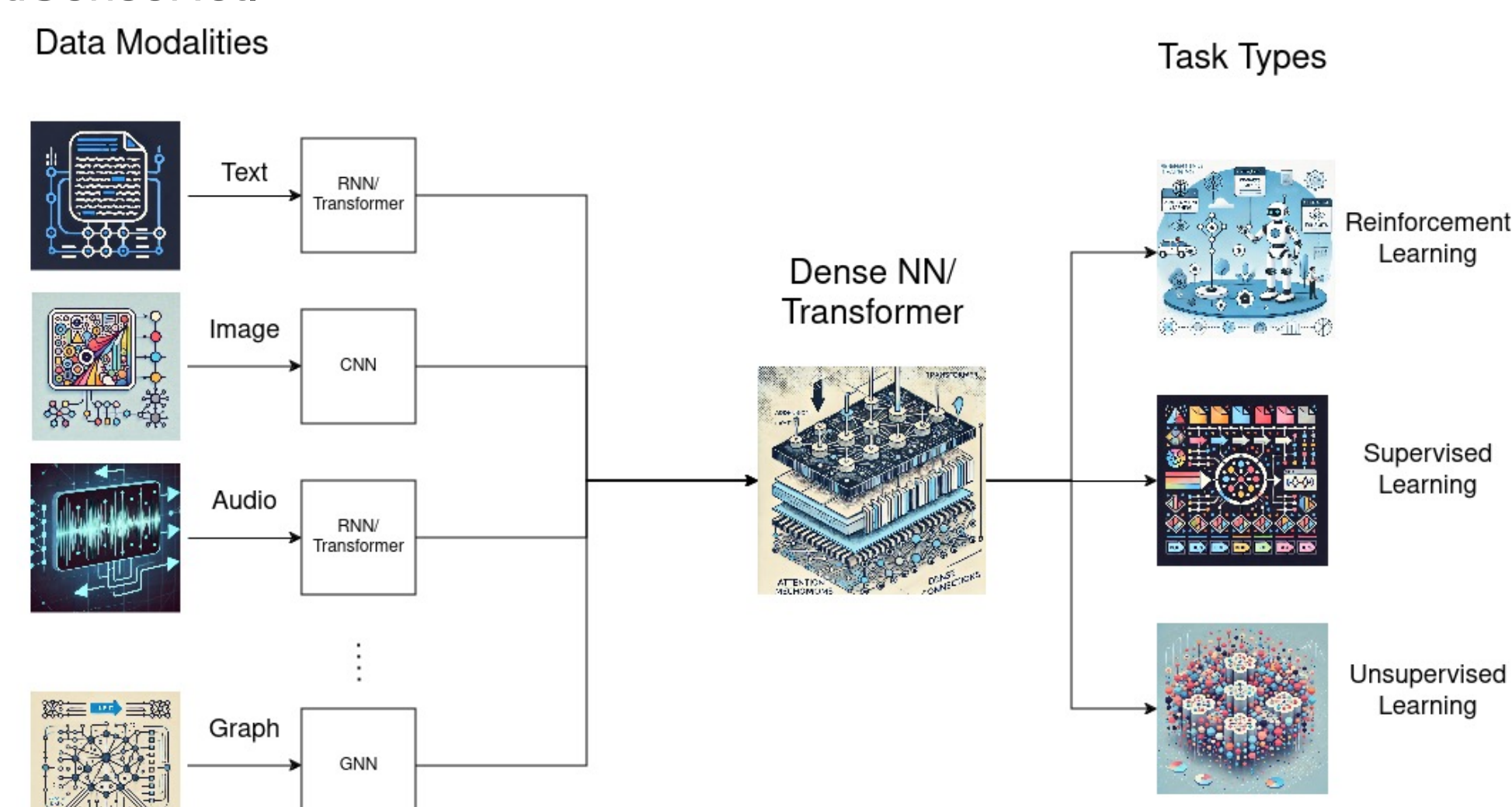
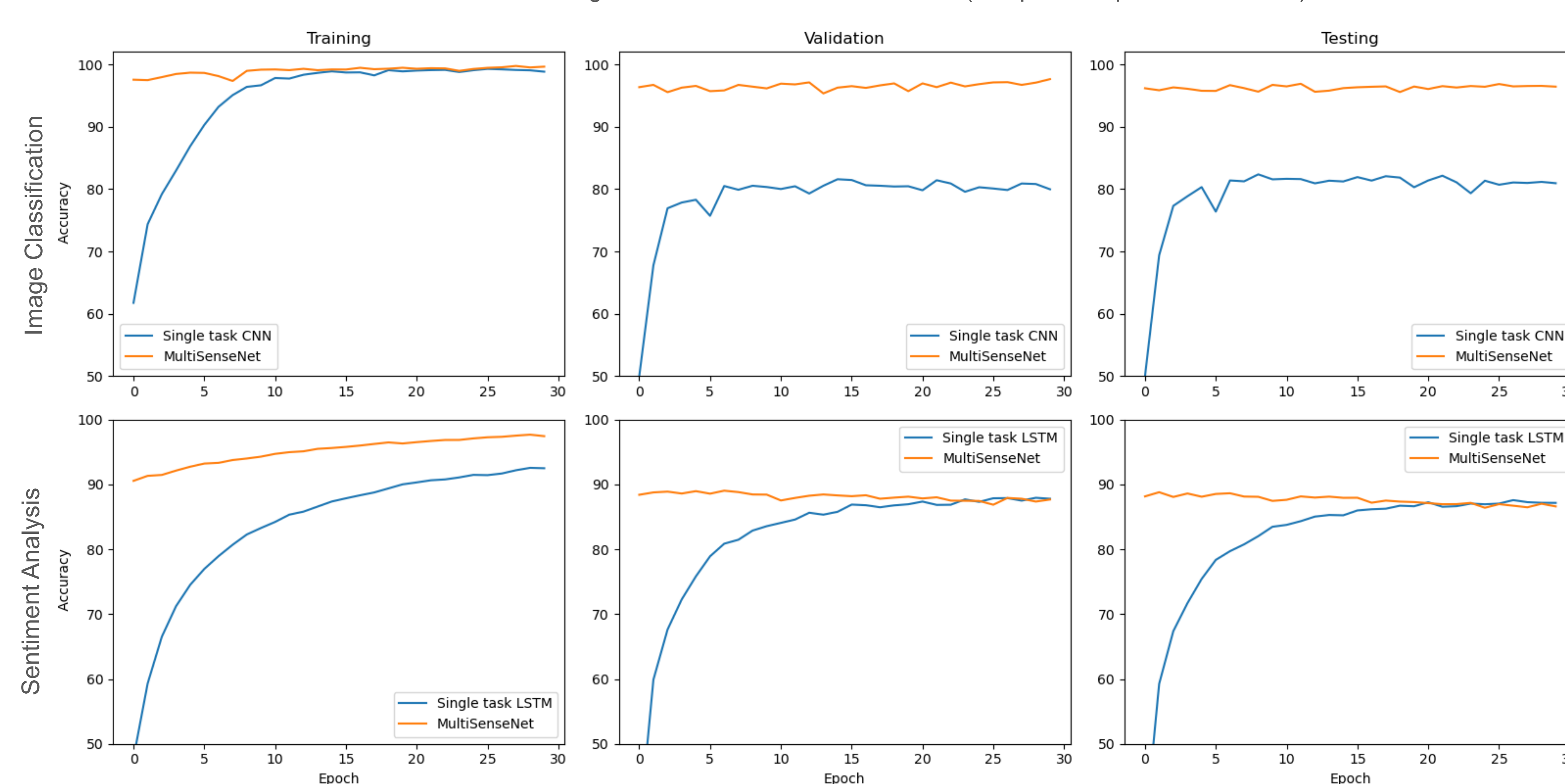


Fig. 2 MultiSenseNet Architecture

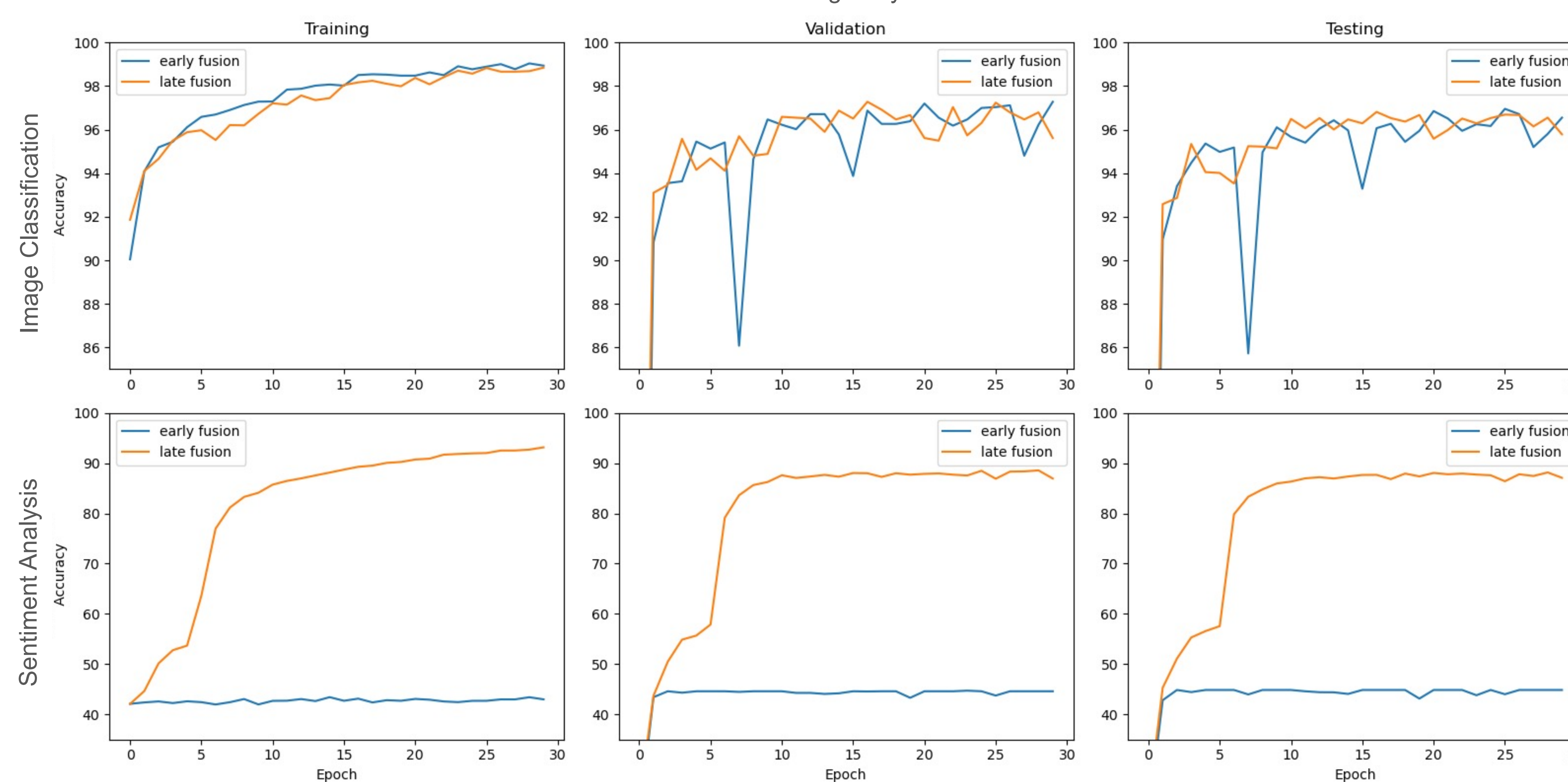
❖ **Training:** So far, we trained five different model architectures on two modalities (image and text) and two subtasks (image classification and sentiment analysis). These include three multimodal models (MSNs) with pre-trained components using early fusion, late fusion, and independent path approaches. The early fusion model combines modalities early, the late fusion model combines them later, and the independent paths model processes each modality separately. Additionally, we built single-task non-pre-trained CNN and LSTM architectures for image classification and sentiment analysis, respectively. Finally, We compared the performance of the fusion methods to each other and the CNN and LSTM models to the independent path model.

RESULTS

Performance of single task models vs MultiSenseNet (independent path architecture)



Performance of MultiSenseNet using early vs late fusion architectures



REFERENCES

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CONCLUSION

❖ MultiSenseNet outperforms the single-task CNN on the image task and also shows a slight improvement compared to the single-task LSTM for the test task. We expect this to be the case because the MSN utilized models that were already pre-trained on vast amounts of data, allowing them to generalize well to new data compared to the single-task models, which were being trained from scratch.

❖ MultiSenseNet performs well on both text and image tasks when using the late fusion compared to the early fusion approach. This is because late fusion preserves more modality-specific features by allowing each modality to be processed independently up to a later stage. In contrast, early fusion combines features from both modalities at an early stage, which can dilute important modality-specific information.

❖ Integrating pre-trained models into MultiSenseNet reduces overall training costs by leveraging learned features, accelerating convergence, and decreasing the need for large datasets and extensive computational resources. This enhances its performance as compared to using non-pre-trained models.

FUTURE WORK

❖ Evaluate the performance of unified representations on held-out test tasks and out-of-distribution datasets.

❖ Implement more data modalities and tasks (unsupervised and reinforcement learning) on MultiSenseNet architecture.

❖ Quantify important characteristics of ML tasks and datasets.

❖ Apply unified framework advancements to climate change-related problem domains, where previous models are performing poorly.

❖ Develop a software solution to automate the transformation of new ML tasks and datasets into our unified framework using Large Language Models (LLMs).

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